

Daniel Foister, Jason Gun, Timyathus Spikes, Tyrese Dickson - DTJT Group
Shehenaz Shaik
CSCI 4350
Sprint #9 - 4/9-4/16

Roles:

- Product Owner: Daniel Foister
- Scrum Master: Timyathus Spikes
- Development Team: Jason Gun, Tyrese Dickson

Time Limit: Two Days (4/9, 4/14, 4/16)

Attendance: All group members were present this week.

Project Github Repo: <https://github.com/12061062/StockBuddy>

Sprint Goal:

- The goal for this sprint was to explore emerging architectures by implementing event-driven functionality, refactoring features into standalone functions, and analyzing how the system could transition to a serverless architecture. These improvements help show how modern architectural approaches could potentially improve StockBuddy.

PBIs/User Stories/To-Do Chosen for Sprint:

- **SCRUM-67:** Trigger specific functions based on user actions (checkout | update inventory | log transaction).
 - Estimation: 4 story points
- **SCRUM-68:** Refactor a small feature into a standalone function to simulate a serverless (FaaS) approach.
 - Estimation: 5 story points
- **SCRUM-69:** Identify parts of the system that could be converted into event-driven functions.
 - Estimation: 4 story points
- **SCRUM-70:** Document how the system could transition to a serverless architecture.
 - Estimation: 4 story points

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- The PBIs/tasks selected for this sprint were based on our team's average velocity of around 16 story points per sprint from previous sprints. These features were prioritized because they introduce emerging architecture concepts such as event-driven design and serverless (FaaS) approaches, while improving the system's flexibility.

Sprint Planning

- During sprint planning, our team reviewed the backlog and selected/created four PBIs/tasks that aligned well with StockBuddy's current development stage. The tasks were distributed evenly among our team members, and the acceptance criteria were clarified before beginning implementation.

Total Sprint Duration:

- 4 members x 8 hours/week x 1 Week/Sprint = **32 man-hours available**

Tasks Performed:

- Task Name: Trigger specific functions based on user actions (checkout | update inventory | log transaction).
- Task Owner: Tyrese Dickson
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: None
- Lessons Learned: Event-driven design improves responsiveness.

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- Task Name: Refactor a small feature into a standalone function to simulate a serverless (FaaS) approach.
- Task Owner: Timyathus Spikes
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: Ran into some problems with system behavior when breaking the existing function into independent components.
- Lessons Learned: None

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- Task Name: Identify parts of the system that could be converted into event-driven functions.
- Task Owner: Jason Gun
- Planned Duration: 3 hours
- Actual Duration: 4 hours
- Status: Completed
- Challenges: None
- Lessons Learned: None

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- Task Name: Document how the system could transition to a serverless architecture.
- Task Owner: Daniel Foister
- Planned Duration: 7 hours
- Actual Duration: 7 hours
- Status: Completed
- Challenges: Had to do a bit of research to understand how we could potentially translate our existing architecture into a serverless model.
- Lessons Learned: Proper and thorough documentation is important for understanding architectural transitions.

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Sprint Review:

- Was the Sprint Goal Achieved?:
 - The Sprint Goal was Achieved.
 - Definition of Done (DoD)
 - Our DoD: All work/code maps to the requirements specified in the Jira Board (backlog) and meets the defined acceptance criteria.

Each completed item has been fully implemented, tested, and verified to function correctly within the system. Code has been reviewed by team members, integrated into the main project without conflicts, and does not introduce new defects. Any necessary documentation has been updated, and the feature is considered complete, stable, and ready for demonstration or release.

- Our team successfully implemented event-driven functionality, refactored features into standalone functions, identified components possible for event-driven architecture, and documented how the system could transition to a serverless architecture. These improvements demonstrate how emerging architectures can improve our system's design and scalability.
- Focus for the Next Sprint:
 - Metrics and Monitoring; Project Wrap-up; Documentation
- PO/Stakeholder Review:
 - The Product Owner reviewed all completed tasks/backlog items:
 - Functions are successfully triggered based on user actions.
 - System components have been modularized into standalone functions.
 - Event-driven opportunities within the system have been identified.
 - Documentation clearly outlines how the system could transition to a serverless architecture.
 - Feedback Provided:
 - Consider further 'refinement' of event-driven functionality.
 - Priority Revisions:
 1. System performance improvements prioritized
- Testing
 - During this sprint, we implemented tests to verify that event-driven functionality and modular components behave correctly. These tests made sure that system actions trigger the correct functions and that refactored components maintain correct behavior.
 - Unit Tests:
 - Test that functions are triggered correctly based on user actions.
 - Test that refactored standalone functions produce correct outputs.

Sprint Retrospective:

- Changes in the Scrum Process:
 - Again, our team does not believe we need to make any changes to the Scrum Process yet, in the context of this Sprint.
- Start: Applying modular and event-driven design principles during development.
- Stop: Building tightly coupled features without considering architectural flexibility.
- Continue: Frequent push requests/commits, clear communication, and consistent task tracking.
- Other Planned Adjustments/Concerns:
 - No major concerns were identified during this sprint.
- Team Velocity:

- Sprint 1: 12 story points
- Sprint 2: 16 story points
- Sprint 3: 18 story points
- Sprint 4: 17 story points
- Sprint 5: 17 story points
- Sprint 6: 16 story points
- Sprint 7: 16 story points
- Sprint 8: 16 story points
- Sprint 9: 17 story points
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- Average Velocity: ~16 story points per sprint

Backlog Grooming

- PBIs/Tasks Added/Removed/Reprioritized
 - As a user, I want a clear and detailed user manual so that I can understand how to use all features of the system effectively.
 - As a developer, I want to perform final testing and improve code quality so that the system is stable, reliable, and ready for final delivery.
 - As a manager, I want to view basic system metrics such as total sales and items sold so that I can monitor business performance.
 - Create the user manual describing how to use StockBuddy.
 - Perform final system testing to ensure all features function correctly.
 - Clean up and refactor code for readability and maintainability.
 - Display simple metrics (total sales, items sold) within the application.
- Definition of Ready (DoR)
 - Our DoR: A backlog item is considered ready when it has clearly defined acceptance criteria, estimated story points, and any dependencies have been identified. The task is well-understood by the team, correctly scoped, and small enough to be completed within a single sprint. All required resources, tools, and information are available, allowing the team to begin work without blockers.

Daily Standup #1:

- Duration: 5 Minutes
- Facilitator: Timyathus Spikes

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Tyrese Dickson

- What did I do yesterday?**
I wrapped up working on hashing and securely storing user credentials instead of using plain text or hardcoded values.
- What do I plan to do today?**
Today, I planned out how I will trigger specific functions based on user actions (checkout | update inventory | log transaction).
- What are any challenges or impediments that I have experienced?**
None

Timyathus Spikes

1. **What did I do yesterday?**

I wrapped up working on sanitizing input fields to reduce the risk of injection attacks.

2. **What do I plan to do today?**

Today, I planned out how I will refactor a small feature into a standalone function to simulate a serverless (FaaS) approach.

3. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

1. **What did I do yesterday?**

I wrapped up working on ensuring sensitive data is not exposed in logs or error messages.

2. **What do I plan to do today?**

Today, I planned out how I will identify parts of the system that could be converted into event-driven functions.

3. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

1. **What did I do yesterday?**

I wrapped up working on reinforcing role-based access control across all system features.

2. **What do I plan to do today?**

Today, I planned out how I will document how the system could transition to a serverless architecture.

3. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #2:

- Duration: 5 Minutes
- Facilitator: Timyathus Spikes

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Tyrese Dickson

4. **What did I do yesterday?**

Yesterday, I planned out how I would trigger specific functions based on user actions (checkout | update inventory | log transaction).

5. **What do I plan to do today?**

Today, I am going to implement the triggering of specific functions based on user actions (checkout | update inventory | log transaction), which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

4. **What did I do yesterday?**

Yesterday, I planned out how I would refactor a small feature into a standalone function to simulate a serverless (FaaS) approach.

5. **What do I plan to do today?**

Today, I am going to refactor a small feature into a standalone function to

simulate a serverless (FaaS) approach, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

4. **What did I do yesterday?**

Yesterday, I planned out how I would identify parts of the system that could be converted into event-driven functions.

5. **What do I plan to do today?**

Today, I am going to identify parts of the system that could be converted into event-driven functions, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

4. **What did I do yesterday?**

Yesterday, I planned out how I would document how the system could transition to a serverless architecture.

5. **What do I plan to do today?**

Today, I am going to document how the system could transition to a serverless architecture, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #3:

- Duration: 5 Minutes
- Facilitator: Timyathus Spikes

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Tyrese Dickson

7. **What did I do yesterday?**

I continued working on implementing the triggering of specific functions based on user actions (checkout | update inventory | log transaction).

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #10.

9. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

7. **What did I do yesterday?**

I continued working on refactoring a small feature into a standalone function to simulate a serverless (FaaS) approach.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #10.

9. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

7. **What did I do yesterday?**

I continued working on identifying parts of the system that could be converted into event-driven functions.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #10.

9. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

7. **What did I do yesterday?**

I continued working on documenting how the system could transition to a serverless architecture.

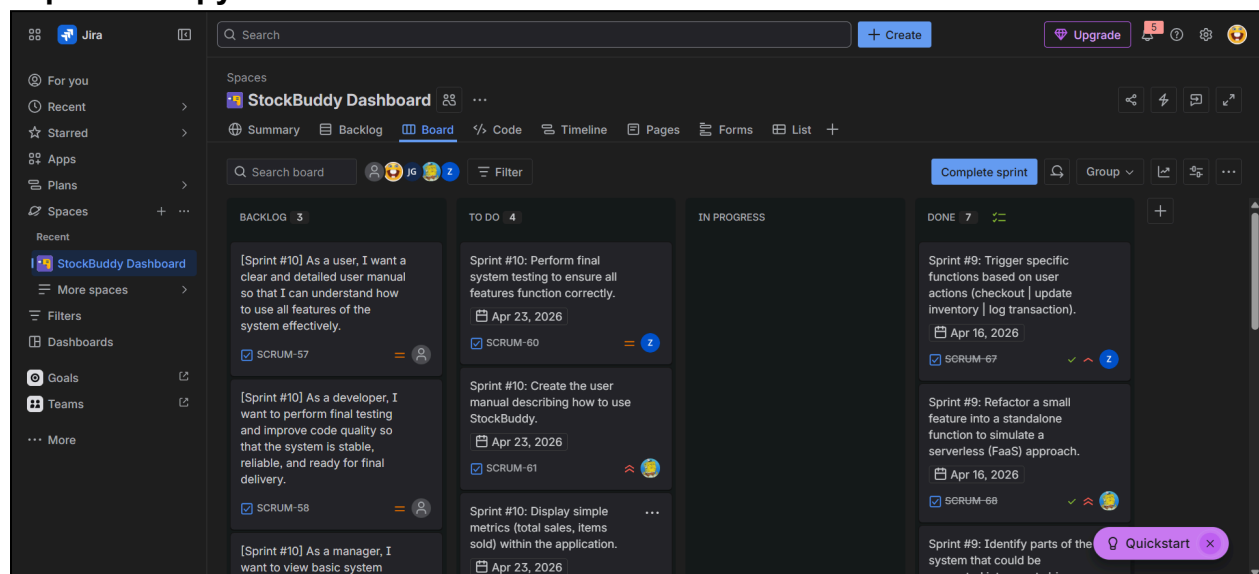
8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #10.

9. **What are any challenges or impediments that I have experienced?**

None

Exported Copy of Scrum Board:



END.